

Performance Testing

Date	02 October 2026
Team ID	6ab7f31f8c7c66e70672939b
Project Name	EduGenie – AI Powered Learning Assistant
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter (HTTP Request samplers, JSON body with text and level)
Type of Testing	Load Testing, Concurrency Testing and Stress Testing
Target Module / API	POST /qa, /quiz, /summarize, /learn/recommendations (Gemini API); POST /explain (local LaMini-Flan-T5-248M)
Test Environment	Local: FastAPI app served by Uvicorn at 127.0.0.1:8000 (single process); JMeter on the same machine; Gemini API over the internet
Test Date	Pending: to be recorded when the JMeter test plan is executed

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Normal load: POST /qa and POST /summarize (Gemini), steady requests with 10 s ramp-up	5	120	HTTP 200 and success=true; avg and max response time compared with Step 3 targets
2	Concurrent load: mixed POST /qa, /quiz, /summarize and /learn/recommendations (Gemini)	10	180	All responses HTTP 200; error rate < 1%; no 504 timeouts
3	Higher AI-processing load: POST /explain on local LaMini-Flan-T5 (4 model generations per request)	3	120	HTTP 200; CPU and memory < 80%; response time recorded

4	Peak / stress load: mixed requests to all five POST endpoints, 60 s ramp-up	25	300	Graceful 429/503/504 handling and model fallback; error rate and limit recorded
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Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	Not measured	Pending	Requires JMeter run; depends on Gemini latency (and local T5 inference for /explain)
2	Response Time (Max)	< 5 seconds	Not measured	Pending	Requires JMeter run; app timeout is 45 s and retry back-off can add up to 8 s
3	Throughput (Req/sec)	-	Not measured	Pending	Requires JMeter run; limited by Gemini quota and CPU
4	Error Rate	< 1%	Not measured	Pending	Requires JMeter run; failures return HTTP 400/502/503/504
5	CPU Utilization	< 80%	Not measured	Pending	Requires OS monitor during the run, mainly for /explain
6	Memory Utilization	< 80%	Not measured	Pending	Requires OS monitor; T5 model stays loaded in the process

Step 4: Observations & Analysis

Key Findings:

No performance test has been executed yet, so no measured result is claimed in this report. Findings below come from review of the source code.

- /qa, /quiz, /summarize and /learn/recommendations are async handlers that await Gemini through client.ai, so the event loop is not blocked while waiting.
- /explain runs the local LaMini-Flan-T5-248M model via asyncio.to_thread and makes four sequential generations per request.
- Failures return JSON errors: 400 invalid input, 502 AI error, 503 busy/not configured, 504 timeout (45 s).

Bottlenecks Identified (expected from code review, not yet observed):

- Gemini latency and quota: on 429/500/503 the app retries twice per model with a 2 s sleep, then falls back from gemini-3.5-flash to gemini-3.1-flash-lite.
- /explain CPU load: concurrent requests share one process and CPU; the model loads lazily on the first call (cold start) without a lock.
- Deployment: single Uvicorn process (documented command: uvicorn main:app --reload); no caching or rate limiting.

Optimization Steps Taken:

None applied yet (no test run). Already in the code: model fallback, retry with back-off, 45 s timeout,

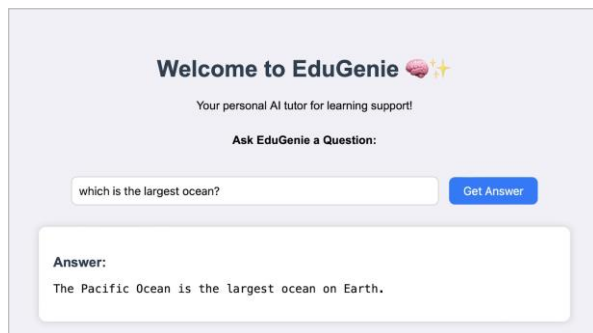
4000-character input limit. To try after the first run: warm up the T5 pipeline at startup, multiple Uvicorn workers, caching repeated prompts.

Note (documentation vs code):

The document mentions Gemini 1.5 Pro, LaMini-Flan-T5-783M and 3 quiz questions. The code uses gemini-3.5-flash (fallback gemini-3.1-flash-lite), LaMini-Flan-T5-248M and 5 questions. This report follows the code.

Step 5: Screenshots / Evidence

JMeter / Locust / k6 output is not available because the performance test has not been executed. The screenshots below are EduGenie application evidence taken from the project documentation, showing the endpoints selected for testing:



Welcome to EduGenie 🧠🌟

Your personal AI tutor for learning support!

Ask EduGenie a Question:

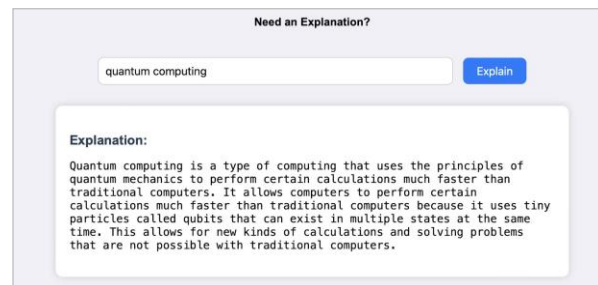
which is the largest ocean?

Get Answer

Answer:

The Pacific Ocean is the largest ocean on Earth.

POST /qa - Question answering



Need an Explanation?

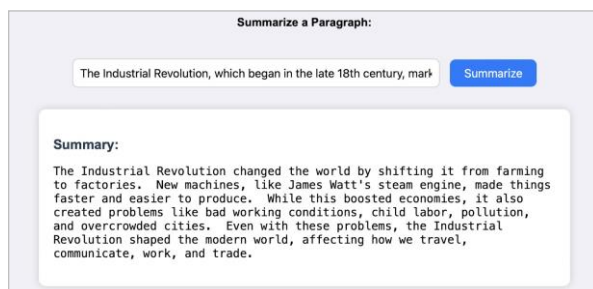
quantum computing

Explain

Explanation:

Quantum computing is a type of computing that uses the principles of quantum mechanics to perform certain calculations much faster than traditional computers. It allows computers to perform certain calculations much faster than traditional computers because it uses tiny particles called qubits that can exist in multiple states at the same time. This allows for new kinds of calculations and solving problems that are not possible with traditional computers.

POST /explain - Concept explanation



Summarize a Paragraph:

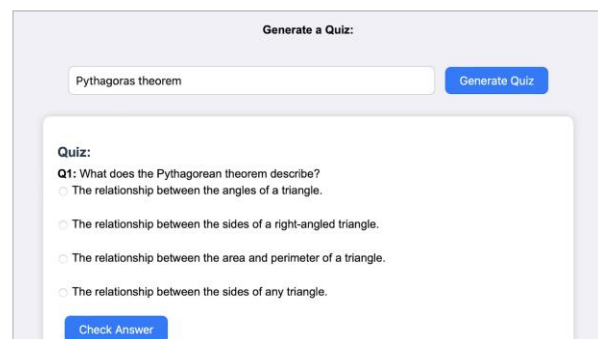
The Industrial Revolution, which began in the late 18th century, marked

Summarize

Summary:

The Industrial Revolution changed the world by shifting it from farming to factories. New machines, like James Watt's steam engine, made things faster and easier to produce. While this boosted economies, it also created problems like bad working conditions, child labor, pollution, and overcrowded cities. Even with these problems, the Industrial Revolution shaped the modern world, affecting how we travel, communicate, work, and trade.

POST /summarize - Summarization



Generate a Quiz:

Pythagoras theorem

Generate Quiz

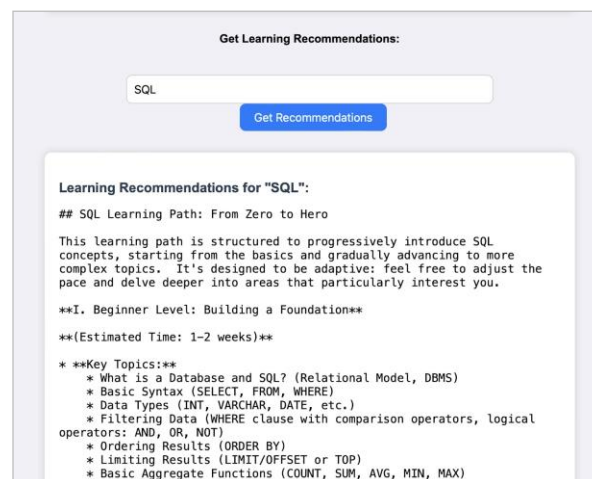
Quiz:

Q1: What does the Pythagorean theorem describe?

- ☐ The relationship between the angles of a triangle.
- ☐ The relationship between the sides of a right-angled triangle.
- ☐ The relationship between the area and perimeter of a triangle.
- ☐ The relationship between the sides of any triangle.

Check Answer

POST /quiz - Quiz generation



Get Learning Recommendations:

SQL

Get Recommendations

Learning Recommendations for "SQL":

SQL Learning Path: From Zero to Hero

This learning path is structured to progressively introduce SQL concepts, starting from the basics and gradually advancing to more complex topics. It's designed to be adaptive: feel free to adjust the pace and delve deeper into areas that particularly interest you.

****I. Beginner Level: Building a Foundation****

****(Estimated Time: 1-2 weeks)****

*** **Key Topics:****

- * What is a Database and SQL? (Relational Model, DBMS)
- * Basic Syntax (SELECT, FROM, WHERE)
- * Data Types (INT, VARCHAR, DATE, etc.)
- * Filtering Data (WHERE clause with comparison operators, logical operators: AND, OR, NOT)
- * Ordering Results (ORDER BY)
- * Limiting Results (LIMIT/OFFSET or TOP)
- * Basic Aggregate Functions (COUNT, SUM, AVG, MIN, MAX)

POST /learn/recommendations - Learning path